ZHANG zihui

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EDUCATION

The Hong Kong Polytechnic University 2026.02–now
PostDoc, Department of Computing *Advisor: Bo Yang*
The Hong Kong Polytechnic University 2021.09–2026.01
PhD, Department of Computing *Advisor: Bo Yang*
Xi'an Jiaotong University 2018.09-2021.06
MSc, School of Mathematics and Statistics *Advisor: Huibin Li*
University of Electronic Science and Technology of China 2014.09-2018.06
BSc, School of Mathematical Science

RESEARCH INTERESTS:

Unsupervised Learning in 3D Vision, 3D Generative Models

PUBLICATIONS:

1. **EvObj: Learning Evolving Object-centric Representations for 3D Instance Segmentation without Scene Supervision**
   *Jiahao Chen, **Zihui Zhang**, Yafei Yang, jinxi Li, Shenxing Wei, Zhixuan Sun, Bo Yang*
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026
2. **GrowSP++: Growing Superpoints and Primitives for Unsupervised 3D Semantic Segmentation**
   ***Zihui Zhang**, Weisheng Dai, Bo Yang, Bing Wang, Bo Li*
   IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2026
3. **GrabS: Generative Embodied Agent for 3D Object Segmentation without Scene Supervision**
   ***Zihui Zhang**, Yafei Yang, Hongtao Wen, Bo Yang*
   International Conference on Learning Representations (**ICLR**), 2025 **(Spotlight Top 5%)**
4. **LogoSP: Local-global Grouping of Superpoints for Unsupervised Semantic Segmentation of 3D Point Clouds**
   ***Zihui Zhang**, Weisheng Dai, Hongtao Wen, Bo Yang*
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025
5. **unMORE: Unsupervised Multi-Object Segmentation via Center-Boundary Reasoning**
   *Yafei Yang, **Zihui Zhang**, Bo Yang*
   The International Conference on Machine Learning (**ICML**), 2025
6. **Boundary-aware Multiclass Spinal Cord Tumor Segmentation on 3D MRI Scans**
   *Hongkang Song, **Zihui Zhang**, Yanpeng Zhou, Jie Hu, Zishuo Wang, Hou Him Chan, Chon Lok Lei, Chen Xu, Yu Xin, and Bo Yang*
   The European Conference on Computer Vision Workshop (**ECCV workshop**), 2024
7. **Adaptive Representation Learning and Sample Weighting for Low-Quality 3D Face Recognition**
   *Cuican Yu, Fengxun Sun, **Zihui Zhang**, Huibin Li, Limin Chen, Jian Sun, Zongben Xu*
   Pattern Recognition (**PR**), 2024
8. **GrowSP: Unsupervised Semantic Segmentation of 3D Point Clouds**
   ***Zihui Zhang**, Bo Yang, Bing Wang, Bo Li*
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023
9. **Meta-learning-based adversarial training for deep 3D face recognition on point clouds**
   *Cuican Yu, **Zihui Zhang**, Huibin Li, Jian Sun, Zongben Xu*
   Pattern Recognition (**PR**), 2023
10. **Learning Flexibly Distributional Representation for Low-quality 3D Face Recognition**
    ***Zihui Zhang**, Cuican Yu, Shuang Xu, Huibin Li*
    The AAAI Conference on Artificial Intelligence (**AAAI**), 2021
11. **Learning Distribution Independent Latent Representation for 3D Face Disentanglement**
    ***Zihui Zhang**, Cuican Yu, Huibin Li, Jian Sun, Feng Liu*

The International Conference on 3D Vision (**3DV**), 2020

12. **A Distribution Independence based Method for 3D Face Shape Decomposition**
    *Cuican Yu, **Zihui Zhang**, Huibin Li, Jian Sun, Zongben Xu*
    Computer Vision and Image Understanding (**CVIU**), 2021

ACADEMIC ACTIVITIES:

Journal Reviewer: IJCV, TNNLS, TCSVT.

Conference Reviewer: NeurIPS, CVPR, ICCV, ICML, ECCV